Rocket Damping

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Repository: `https://github.com/sidd-somadder/rogue-fin-design`

Per the FAR-OUT rules, the damping ratio ζ of the rocket must lie between 0.03 and 0.5, and the association recommends a design target of 0.05 to 0.3. The dynamic stability of the rocket must hold for wind speeds up to 30 mph [3].

While OpenRocket lets us add in simulation listeners and custom expressions for damping, there was an inconsistency with the damping moment coefficient. To expedite hand calculations for the damping ratio as done in iteration 2, It was decided that a tool should be built using flight simulation data from OpenRocket. The primary driver for this decision is due to the damping moment listener inconsistency, and it would be easier to check damping from different planforms just by importing simulation data rather than checking each time. See `damping.py` in repository.

First, some preliminary research was done on dynamic stability. The main text used was Mandell's *Topics in Advanced Model Rocketry (1973)*. The corrective moment (M_c) and damping moment (M_d) are best understood from modeling a rocket's angle as a damped oscillator. Namely, the corrective moment coefficient and damped moment coefficients are linear approximations in that they are linearly proportional to the torques felt by the rocket by the angle of attack and angular velocity, respectively:

$$\frac{d^2\alpha}{dt^2} + C_{dm}\frac{d\alpha}{dt} + C_{cm}\alpha = 0$$

$$C_{cm} = \frac{1}{I_L}\frac{dM_c}{d\alpha}\Big|_{\alpha=0} \qquad C_{dm} = \frac{1}{I_L}\frac{dM_d}{d\omega}\Big|_{\omega=0}$$

It can be shown that these coefficients may be computed from the slope of the coefficient of normal force w.r.t. angle of attack ($C_{N\alpha}$) per each main rocket component. As a simplification, normal force effects on the body tube are neglected, and since the Rogue rocket has no shoulder/transition, the two components are the nose cone and fins.

Following Mandell [1], the damping ratio is

$$\zeta = \frac{C_{dm}}{2\sqrt{C_{cm}\, I_L}}$$

where C_{cm} is the corrective moment coefficient (lbf ft/rad), C_{dm} is the damping moment coefficient (lbf ft s/rad), and I_L is the longitudinal moment of inertia ($slug \cdot ft^2$) about the CG (ft). Written as sums over the components i (nose cone and fin set),

$$C_{cm} = \tfrac{1}{2}\rho V^2 A_{ref} \sum_i C_{N\alpha,i}\,(X_i - X_{cg})$$

$$C_{dm} = \tfrac{1}{2}\rho V A_{ref} \sum_i C_{N\alpha,i}\,(X_i - X_{cg})^2 \;+\; \dot{m}\,(L_{ne} - X_{cg})^2$$

where X_i is the center of pressure of component i and X_{cg} the center of gravity, both measured from the nose tip, and $A_{ref} = \pi d^2/4$ with d the body diameter. In C_{dm} the arms are squared, so both the nose cone and fins contribute to damping. The second term of C_{dm} is jet damping, with L_{ne} the distance from the nose tip to the nozzle exit.

Normal force coefficient slopes. For a sharp nose cone whose base diameter equals the reference diameter, $(C_{N\alpha})_N = 2$ per radian, independent of shape and Mach number. For a set of $N = 3$ or 4 fins, Mandell gives [1, Eq. 86]

$$(C_{N\alpha})_F = \frac{N\,AR\left(\dfrac{C_r + C_t}{2r_t}\right)\left(\dfrac{s}{r_t}\right)}{2 + \sqrt{4 + \left(\dfrac{\beta\,AR}{\cos\Gamma_c}\right)^2}}, \qquad \tan\Gamma_c = \frac{C_r - C_t}{2s}$$

where C_r and C_t are the root and tip chords, s the fin span, r_t the body radius, and Γ_c the sweep of the mid-chord line (the expression for Γ_c assumes a clipped delta with an unswept trailing edge). The factor β is the compressibility correction discussed below; $\beta = 1$ recovers Mandell's equation. The $(C_{N\alpha})_F$ is multiplied with the interference coefficient K to include the body tube's influence [1, Eqs. 87–88]

$$\tau = \frac{s + r_t}{r_t}, \qquad K = 1 + \frac{1}{\tau}, \qquad (C_{N\alpha})_{F,corrected} = K\,(C_{N\alpha})_F$$

with r_t the body radius at the fins ($r_t = d/2$ since our airframe has a constant body diameter).

Aspect ratio. Mandell's Fig. 36 defines the aspect ratio of a single fin as $2s/(C_r + C_t)$. Used in Eq. 86, this value under-predicts the fin normal force. Eq. 85 is term for term the single-fin equation used by OpenRocket [2] when AR is the aspect ratio of the fin together with its mirror image in the body surface:

$$AR = \frac{(2s)^2}{2A_{fin}} = \frac{2s^2}{A_{fin}} = \frac{4s}{C_r + C_t}$$

Compressibility. The full-scale rocket reaches $M \approx 0.67$, so a subsonic compressibility correction is applied to the fins through the Prandtl–Glauert factor

$$\beta = \sqrt{1 - M^2},$$

evaluated at each time step from OpenRocket's Mach number. Following OpenRocket [2], β multiplies the aspect-ratio term inside the square root: compressibility makes the fin behave as if its aspect ratio were βAR. The nose cone slope is a slender-body result and is not corrected. Because a larger fin slope raises both C_{cm} and C_{dm}, the effect on ζ is small; it matters more for the static margin near maximum velocity.

Centers of pressure. For a tangent ogive nose cone of length L_N,

$$X_N = 0.466\,L_N.$$

For the fin set, using the notation of Mandell's Fig. 36, X_T is the distance from the nose tip to the root leading edge (Z_T in Mandell). With the fin trailing edge flush with the aft end of the airframe (length L), $X_T = L - C_r$, and for a clipped delta the leading edge sweeps back by $X_R = C_r - C_t$ over the span. The fin set center of pressure is then

$$\bar{X}_F = X_T + \frac{X_R}{3}\,\frac{C_r + 2C_t}{C_r + C_t} + \frac{1}{6}\left[(C_r + C_t) - \frac{C_r C_t}{C_r + C_t}\right].$$

The interference coefficient scales the fin normal force but does not move its center of pressure. This subsonic location is held fixed; OpenRocket moves the fin center of pressure aft above $M \approx 0.5$, so for full-scale the difference should be checked against OpenRocket's component analysis.

Flight data and units. The velocity V, air density ρ, center of gravity X_{cg}, longitudinal moment of inertia I_L, mass m and Mach number are taken from the OpenRocket simulation at each time step. All quantities are converted to slug–ft–s units before computing ζ: lengths in ft, A_{ref} in ft^2, I_L from $\mathrm{lb_m\,ft^2}$ to $\mathrm{slug\,ft^2}$, mass from oz to slug, and ρ to $\mathrm{slug/ft^3}$. The mass flow rate is estimated from the mass history as $\dot{m} = -\,dm/dt$ by central differences, and the nozzle exit is taken at the aft end of the airframe, $L_{ne} = L$.

Evaluation. ζ is evaluated from rail exit to apogee; before rail exit the launch rail constrains the rocket, and after apogee the rocket descends under recovery. The competition upper and lower limits are checked:

- **Upper limit (0.5)** The damping ratio is highest near rail exit when the motor is burning and the airspeed is lowest. The jet damping contribution, $\dot{m}(L_{ne} - X_{cg})^2/\left(2\sqrt{C_{cm} I_L}\right)$, scales as $1/V$ and is largest here.

- **Lower limit (0.03)** After burnout $\dot{m} = 0$ and V cancels:

$$\zeta_{coast} = \frac{\sqrt{\frac{1}{2}\rho A_{ref}}\,\sum_i C_{N\alpha,i}(X_i - X_{cg})^2}{2\sqrt{I_L \sum_i C_{N\alpha,i}(X_i - X_{cg})}}$$

  so ζ depends only on the air density, geometry, CG and I_L. It decreases slowly with altitude and reaches its minimum near apogee.

The method is linear and assumes small angles of attack. At rail exit in the 30 mph design wind the angle of attack is large (about 20° for the subscale), so the rail-exit value is an estimate.

Verification & Notable Bugs

Some early runs had a largely overestimated damping ratio $\zeta \approx 0.9$ due to accidentally swapping the mathematical definitions of each coefficient during implementation. Afterwards, when testing the flight data importing ability, the damping ratio was largely underestimated, and it was found that the bug came from input data using density in oz/in^3 when it should have been in slug/ft^3.

Two more unit conversions were needed on the imported data. OpenRocket exports the longitudinal moment of inertia in lb·ft^2 (pound-mass), so it is divided by $g = 32.174$ to give slug·ft^2. Mass is exported in ounces, so it is converted to slugs before the jet-damping mass flow rate $\dot{m} = -dm/dt$ is taken with `np.gradient` against the time column; leaving mass in ounces made the jet term roughly 16 times too large.

Before trusting the tool, the fin normal-force slope was checked against OpenRocket. Mandell's Fig. 36 defines the aspect ratio of a single fin, $AR = 2s/(C_r + C_t)$, but with that definition the subscale 3×3×4 fins gave a total $C_{N\alpha}$ (nose and fins) of 5.05, against OpenRocket's 7.1. Initially, it was guessed there was an error with the fin aspect ratio, as it was unusual that Mandell reported it in terms of semi-span while aircraft wings typically use the full tip-to-tip span. As a test, the full span was used, namely $AR = 4s/(C_r+C_t)$, which gave a total of 7.04: 5.04 from the fins and 2 from the nose, without compressibility. OpenRocket's component analysis is evaluated at $M = 0.3$, where the compressibility correction raises the fins to $C_{N\alpha} = 5.131$ and the total to 7.13, matching OpenRocket.

References

[1] G. K. Mandell, G. J. Caporaso, and W. P. Bengen, *Topics in Advanced Model Rocketry*. Cambridge, MA: MIT Press, 1973.

[2] S. Niskanen, *OpenRocket Technical Documentation*, version 13.05, 2013.

[3] Friends of Amateur Rocketry, *FAR-OUT: Friends of Amateur Rocketry – Oxidizers Uninhibited Tournament, 2026-27 Official Competition Rules and Requirements*, Rev. 2026-08-31, Aug. 2026. [Online]. Available: `https://static1.squarespace.com/static/64cc93ec537b6a2fa74f96ba/t/6aa1ff516daf985f5ebb7ae4/1789001554847/FAR-OUT_Rules_Requirements_2026-09-09.pdf` (accessed Oct. 10, 2026).